

Housing Price Prediction — Findings Summary

Regression analysis of 545 properties · 12 features · target: sale price

Objective & Approach

The goal was to predict house prices from property attributes and identify which features drive value. The data (545 rows, no missing values or duplicates) was cleaned, yes/no fields were encoded to 1/0, and furnishing status was one-hot encoded. Two models were trained on an 80/20 train-test split: Linear Regression and a Random Forest Regressor.

Model Performance

On the held-out test set, the simpler Linear Regression model performed best:

Model	MAE	RMSE	R ² Score
Linear Regression ✓ best	₹9,70,043	₹13,24,507	0.653
Random Forest	₹10,14,947	₹13,99,769	0.612

In plain terms, the best model explains about 65% of the variation in price and is off by roughly ₹9.7 lakh on average — a solid ballpark estimator, but not precise enough to set a final listing price alone.

Which Features Matter Most

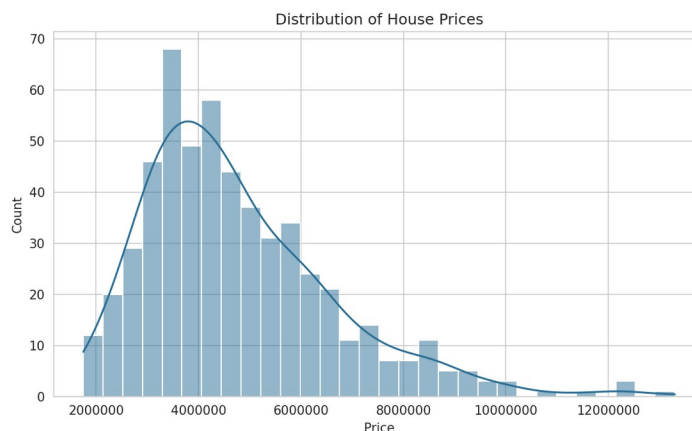
- **Area (size):** the single strongest driver of price (correlation 0.54).
- **Bathrooms:** nearly as influential as area (0.52) — more so than bedrooms.
- **Air conditioning & stories:** both meaningfully lift price (0.45 and 0.42).
- **Parking and preferred-area location:** moderate positive effect; unfurnished homes sell for less.

What Stood Out

- **Simple beat complex:** Linear Regression outperformed Random Forest — the dataset is small and relationships are largely linear, so the flexible model overfit.
- **Bathrooms ≈ area:** bathroom count rivalled total size as a price predictor, which was unexpected.
- **Premium homes under-predicted:** both models undershoot the most expensive properties, which are rare in the data.

Recommendation

Use the model as a fast first-pass valuation tool and in listings emphasise the features that move price most — floor area, bathroom count, and air conditioning. Because the model under-predicts high-end homes, premium properties should still be priced with a human expert rather than the model alone.



Price distribution — right-skewed; most homes fall between ₹3M and ₹6M.